General Maintenance



Shut Down

Careful shut down ensures that the machine can be restarted safely and quickly. These shut down procedures also ensure the best productivity from the machine.

a. Press  on the Operator Station. This allows the machine to finish the current cycle but not continue into further cycles. It will shut down the extruder motor and the clamp.

b. Toggle SETBACK TEMPERATURE to ENABLED on the Temperatures screen.

c. Wait about 15 minutes. Cooling plastic will try to move extruder slowly back. Leaving hydraulics On will hold the screw full forward.

d. Press Control Power to turn Off the Operator Console.

e. After temperatures have cooled below 300° F (150°C) turn main power disconnect Off.

f. Turn water and air Off.

Cool Down

Temperature setback allows the machine to cool for maintenance or other operations. Use setback temperature only when the machine will be left for more than 45 to 60 minutes or when a maintenance procedure will take longer than 45 to 60 minutes to complete. Temperature setback typically reduces machine temperature to approximately 121°C (250°F).

a. Touch  on the menu bar.

b. Select temperature zone.

c. Toggle SETBACK TEMPERATURE to ENABLED on the Temperatures screen.

Note: *SETBACK TEMPERATURES ENABLED is indicated by a background that flashes red and yellow.*

d. Toggle SETBACK TEMPERATURE to DISABLED and wait for temperature to return to normal.

Quick Guide to Cold Start and Cycle Start Procedures

Cold Start Procedure

a. Turn on main power, water, air and downstream equipment.

b. Toggle “user” to UNILOY and enter password.

c. Turn on control power (push the button  and external circuit for downstream equipment ().

d. Turn on heats (). Toggle SETBACK for one hour.

e. Toggle SETBACK off. Start hydraulic pump to warm oil (). Wait for one hour.

f. After heats are off SETBACK for one hour, check temperatures. If temperatures are normal proceed to step #7.

g. Turn off hydraulic system () and clean plastic off dies.

h. Turn on hydraulic pump () and programmer ().

i. Open and close clamp 10 times to warm up the oil ( и ). Raise and lower swing arm 10 times (). Push and release charge button 10 times ().

j. Set extruder RPM to 20. Start the extruder (). When parisons reach the floor increase RPM to production speed minus 10 RPM. Wait for all air to escape from the extruder.

k. Hold charge button () until shot size light () is on. Release charge button to drop parisons. Drop parisons 5 times.

Machine is ready for CYCLE START PROCEDURE.

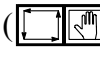
Cycle Start Procedure

a. Make sure that all downstream equipment is on.

b. Start the hydraulic system () and parison programmer ().

c. Swich set up key to 0 (). Bottle divert to 1. And bottle blow off to 1.

d. Move swing arm down ().

e. Move manual/auto switch to AUTO () on both sides of machine.

f. Set extruder speed to 10 RPM below production speed.

g. Start the extruder ().

h. Hold charge button () until shot size light () is on. Release charge button - parisons will drop.

i. Repeat step ‘h’ five times.

j. After dropping five shots push cycle start button ()

k. The bottle tails will be long at first. After about 10 cycles the tails will shorten. When the tails are ok monitor IDLE TIME on the PRODUCTION DATA page. Increase the extruder RPM until IDLE TIME equals 0.0 - 0.2. If IDLE TIME is negative reduce the extruder RPM.

Tooling and Mold Setup

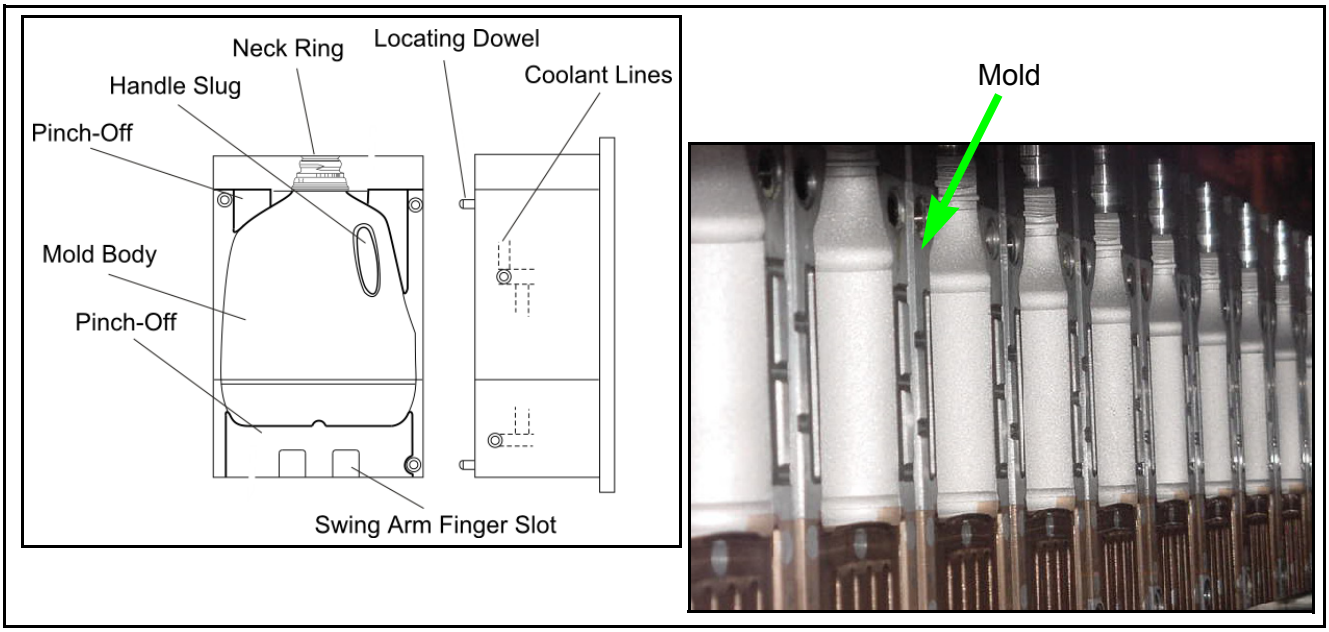
When changing the container or part being produced, follow the instructions in this section to set up the Blow Molding Machine.

This chapter provides step by step setup guidance for technicians trained in the maintenance and operation of the machine. Mold installation and setup should be performed by trained, qualified personnel only. For training information, contact the Uniloy Customer Service Department at (517) 424-8752 or (517) 424-8751, or fax (517) 424-8769.

Setting up of the machine is best accomplished by two people. Setup instructions are broken down into the following general areas:

- Setting Up Molds
- Setting up Stripper Mechanism (Optional)
- Setting Mold Cycle Timers
- Setting Swing Arm Timing (Optional)

Note: *Some features are optional and may not be configured on all machines.*



Molds

Note: *Depending on the model, some molds are mounted to a common backplate that does not require individual hoses. The illustrations in this chapter do not show the maximum number of molds that can be installed.*

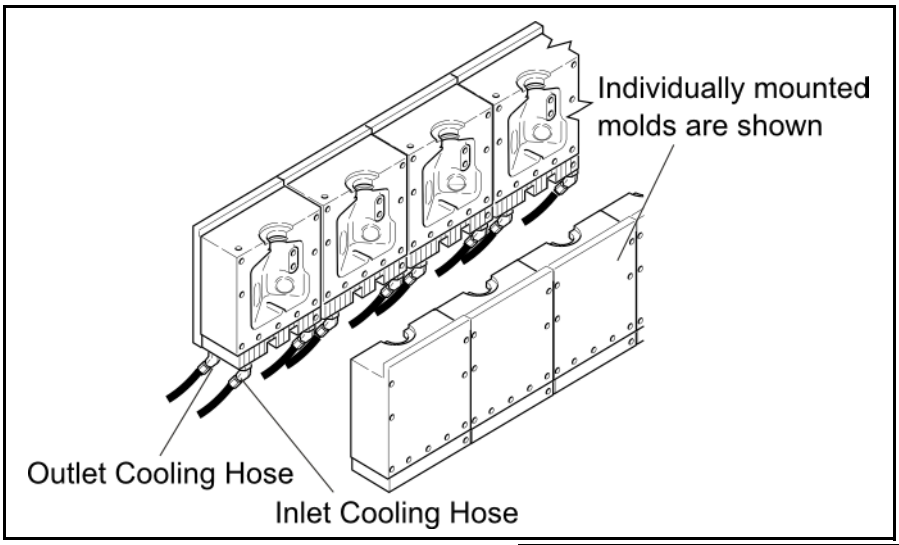
Removing the Molds with Individual Cooling Hoses

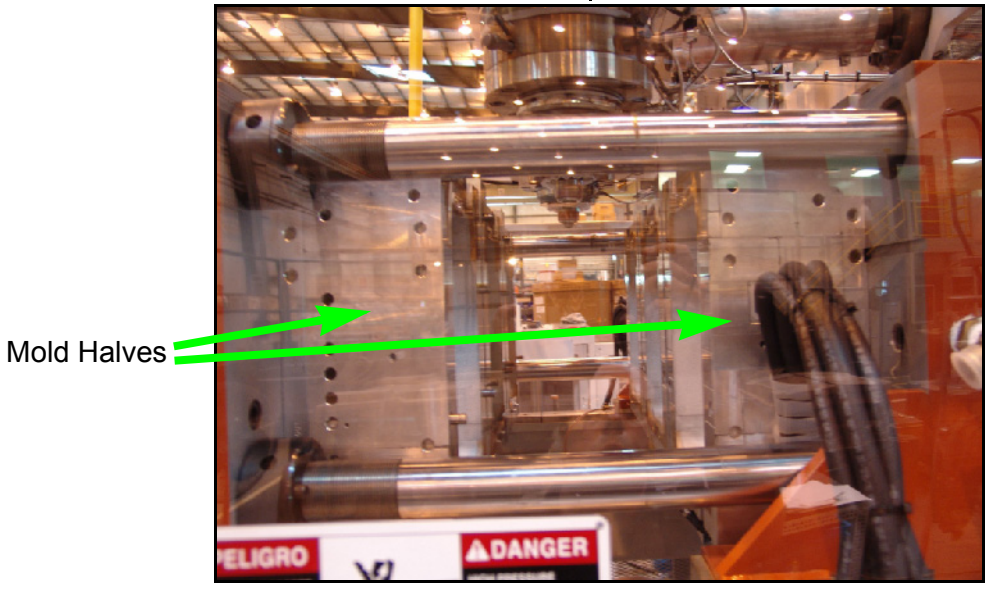
Follow the procedures in this section to remove molds from the machine. Mold halves are not interchangeable. Start from the outside and work inward. Remove one mold at a time and tag each half of the old set.

a. Ensure the machine, chiller, and shut off valves are locked/tagged out.

b. Disconnect the inlet and outlet cooling hoses from fittings. Set the hoses aside for later use.

c. Remove the chute.





Mold Halves

d. Loosen the two lower bolts securing the rear mold half to the platen. Remove the two upper bolts.

e. Slide the mold half off the lower bolts and carefully set it aside.

f. Remove the remaining two bolts from the platen and set them aside.

g. Repeat Steps ‘d’ through ‘f’ for the matching front mold half.

Note: *Keep track of all removed parts and hardware. They will be reused when installing the next set of molds. Protect all parts from damage.*

h. Tag the two mold halves as a matched set. Record the die position on the tag.

⚠ CAUTION

As precision assemblies, molds are easily damaged. Place cardboard between the mold halves and around each mold before transporting to storage. Keep the halves of each mold together. Mold halves are not interchangeable between mold sets.

i. Repeat Steps ‘b’ through ‘h’ for each of the remaining molds.

Removing a Banked Mold

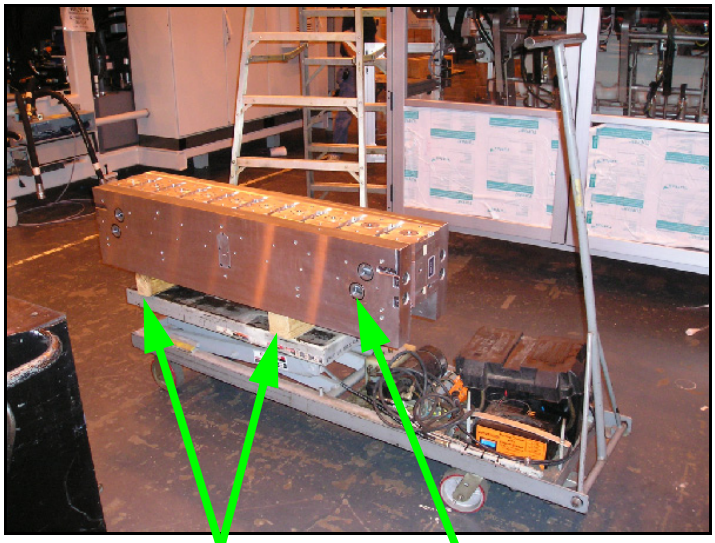
Due to the weight of the Bank Mold, it is best to use a mold cart or similar lifting device. A mold cart can be purchased through Uniloy.

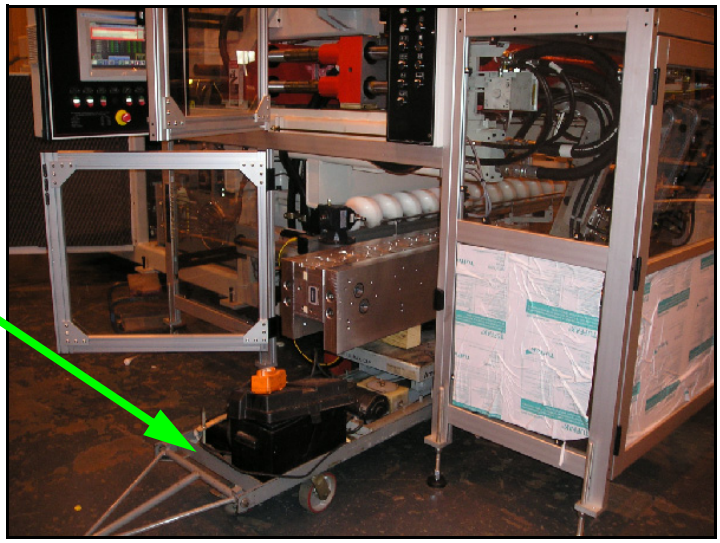
Use the following procedure to remove a Banked Mold:

a. Close the clamp, but do not build tonnage.

b. Ensure the machine, chiller, and shut off valves are locked/tagged out.

c. Position the mold cart under the mold. Wood blocks are typically needed to reach up between the platens to reach the mold.





d. Raise the mold cart until it just makes contact with the mold. DO NOT apply pressure to the mold.

e. Remove the bolts from both halves of the mold.

Installing Molds in the Machine

Mold Alignment

Typically, bank molds are used, and are pre-aligned with the clamp by "keys" for a specific head setter (the centerline of one head or cavity to the next head or cavity). The clamp should be aligned first to the head, then the molds will be aligned by using the "keys." If the mold stack height changes, the clamp must be re-centered to the head.

For Bank Molds

a. Switch the key on the Operator's Console to the Mold Setup position.

b. Support the bottom of the molds with a cart or other approved device, so the molds can be easily positioned onto the keys on the clamp.

c. Carefully align the bank molds to the inside edge of the vertical key and the top edge of the horizontal key.

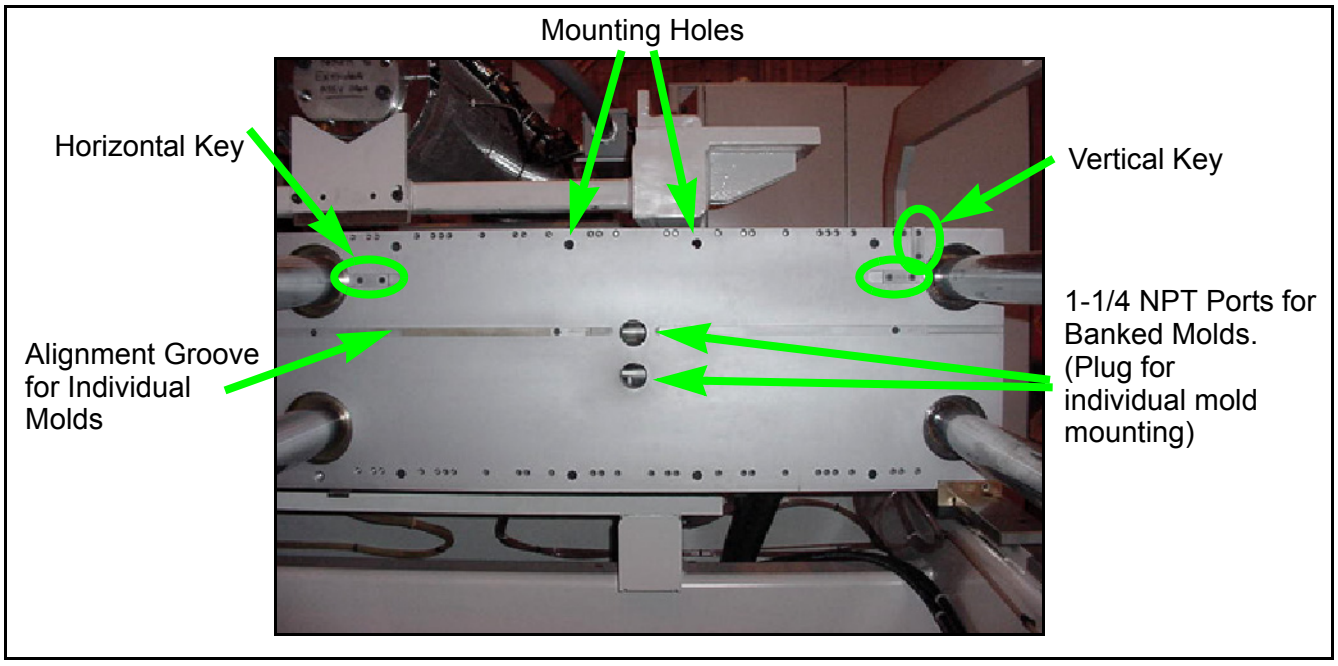
d. When the bank molds are in position on the keys, securely bolt the molds onto the clamp. Make sure the top bolts of each mold bank are tight on both clamps.

For Individual Molds

a. Use the Alignment Groove on the clamp to position each mold vertically.

b. The horizontal position of the mold should be centered with the center of the corresponding head.

c. After the molds have been properly centered, securely bolt the molds onto the clamp. Make sure the top bolts of each mold bank are tight on both clamps.



Mold Alignment

Premature Wear

Premature wear can occur when the alignment pins are not properly seating into the bushings or a mismatch of the molds is creating an uneven parting line in the part. A clicking sound can also be heard when the mold closes if the pins are out of alignment.

It is important to periodically check mold alignment, especially the first time the molds are installed.

Connect Mold Cooling Water Lines

The cooling lines are plumbed to each platen. Banked Molds are fed directly through the platens and no hoses are required.

⚠ CAUTION

To prevent hose failure, do not allow the swing arm to contact or pinch any hose assembly during operation.

Inlet and outlet fittings are located at the bottom of the mold halves when the individual molds are installed on the platens. The inlet fitting is closest to the mold face; the outlet fitting is closest to the platen.

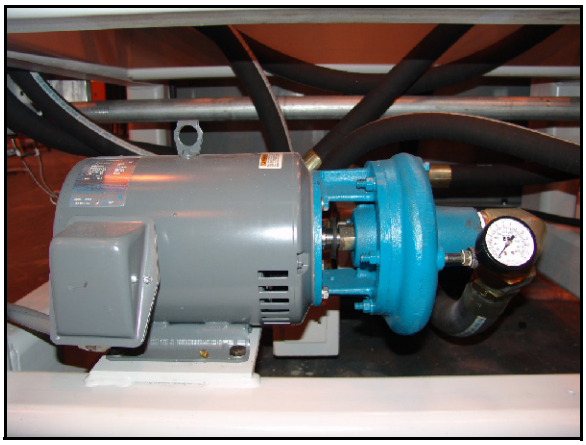


Mold Water Cooling System



Mold Water Cooling Manifold

Water Cooling Lines (R2000 shown)



Mold Water Cooling Pump

Setting Up the Stripper Plate (Optional)

Use the following procedure to install and adjust the stripper plates.

a. Attach the stripper plate loosely to the stripper arms with two screws.

b. Center the circular opening in the stripper plate on the die.

c. Tighten the stripper plate screws.

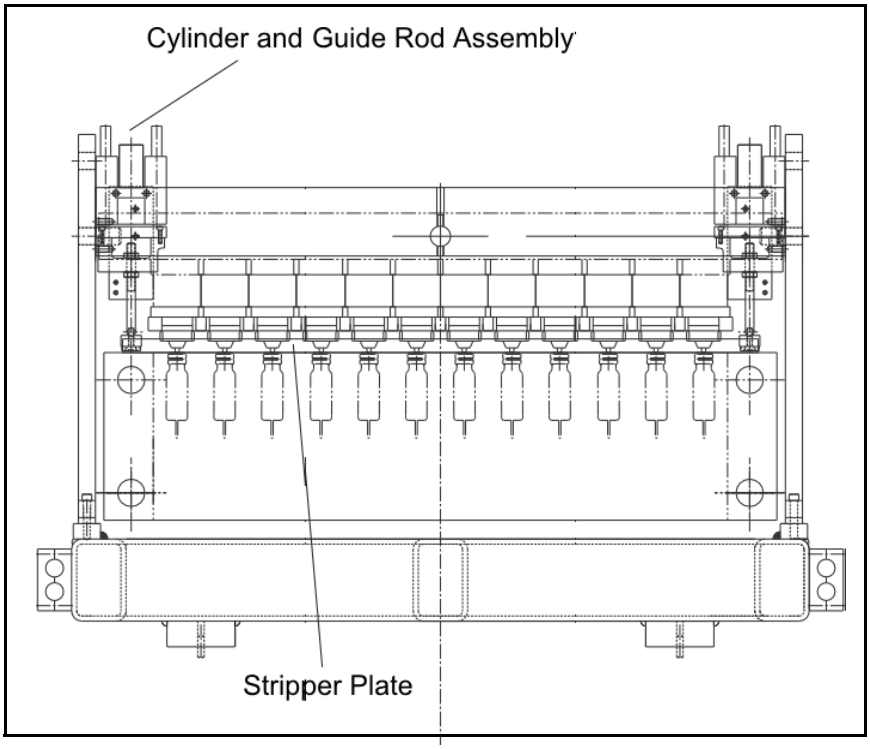
d. Adjust the locknuts on each screw to secure the bottom of the stripper plate at its highest position.

e. Measure the distance from the bottom of the stripper plate to the bottom of the die.

f. Rotate the stripper air cylinder male rod to adjust the stripper plate height. The bottom of the stripper plate should be 1.58mm (.063 in.) above the bottom of the die.

g. Tighten the locking nut on the stripper air cylinder male rod.

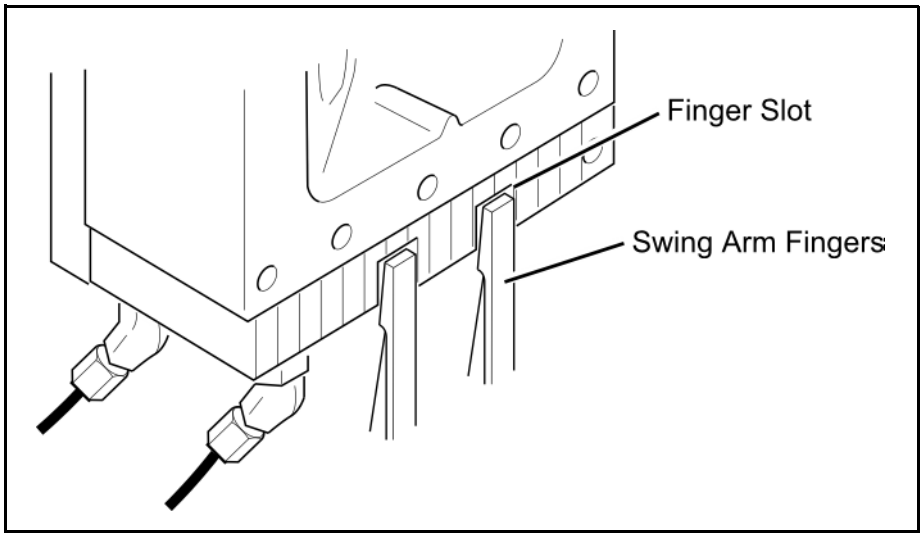
h. Repeat steps ‘a’ through ‘g’ for each of the remaining stripper plates.



Stripper (Optional)

Position Swing Arms (Optional)

Follow the procedures in this section to adjust the swing arms and fingers to match the position of the installed mold halves. Proper adjustment of the swing arms ensures smooth, trouble-free takeout of the blown containers.



Position Swing Arms

a. Position the swing arms in the up position using  on the Operator Station.

b. Check the position of the fingers in the finger slot at the base of each mold. The fingers should be centered in the finger slot and extend to the bottom of the slot without contact. If the fingers are not in position:

   - Raise or lower swing arms. Loosen the four bolts that attach the swing arm to the shaft. Move swing arms into the proper position. Retighten the four bolts.
   - Adjust fingers left or right. Loosen the cap screws at the bottom of each finger and retighten when in the proper position.
   - Adjust fingers front to back. Loosen the four bolts on the slotted mounting bracket for the rotary actuator that moves the swing arm. Retighten bolts when in the proper position.

c. Adjust to close squarely. Loosen the Allen head locking screw, rotate the fingers into position, and retighten the locking screw.

d. Open the clamp.

On certain machines swing arm adjustment can be made to accommodate differences in mold shut height and mold height.

Setting Swing Arm Timing (Optional)

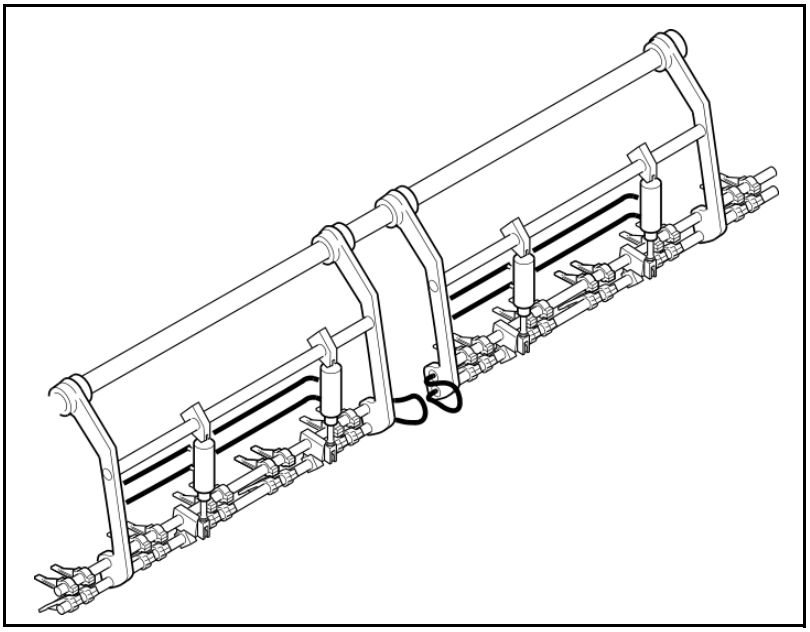
The swing arm mechanism removes the blown container from the mold. Fingers at the end of the swing arm grasp the tail of the blown container.

⚠ CAUTION

If the fingers close too soon, the hot plastic will be pushed up into the molds, requiring the operator to stop the machine to clear them.

Follow the procedures in this section to adjust swing arm timing:

- After installing new molds
- Whenever machine cycle time is adjusted.



Swing Arm

Adjust Swing Arm Hydraulic Pressure and Speed (Optional)

Typically the pressure is set at the factory (refer to the hydraulic print for additional information).

Follow this procedure, in this section, to find the minimum pressure to operate the swing arm. Sufficient pressure is required to ensure smooth actuation of the swing arm mechanism.

a. Turn  Off.

b. Fully open the two flow control valves (counterclockwise) for the swing arm rotary actuator.

c. Turn the swing arm pressure reducing valve counterclockwise to the full open position.

d. Turn  On.

e. Turn  to Up.

Note: *The swing arms should not move when the swing arm switch is turned from Auto to Up.*

f. Slowly close the pressure reducing valve (clockwise) until the swing arms reach the full Up position consistently.

g. Check the pressure reducing valve. It should be possible to stop the swing arms when the swing arm switch is turned from Auto to Up while the hydraulic pump is running.

Note: *The pressure reducing valve requires no further adjustment once the desired result has been achieved.*

h. The swing arm is proportionally controlled, speeds and positions are set via the control. On the B&R control, the settings are found on the Position Control screen under the Swing Arm tab. On the Allen Bradley control, the settings are found in the Position Calibration screen.

Setting the Mold Timing

All setpoints are preset to meet customer specifications before delivery. These setpoints may need adjusting to meet customer site conditions if materials or specifications change or problems occur. Following the procedures in this section may alter the mold cycle timing:

- After installing new molds
- Whenever machine cycle time is adjusted.

Timers in the microprocessor control the mold timing.

Setting Mold Close Pressure

Follow the procedure in this section to set mold close pressure. Correct pressure setup helps ensure product quality.

a. Touch  on the main Menu.

b. Touch MACHINE SETUP - MACHINE ENABLES on the Machine Setup screen.

c. Toggle HIGH PRESSURE to ENABLED.

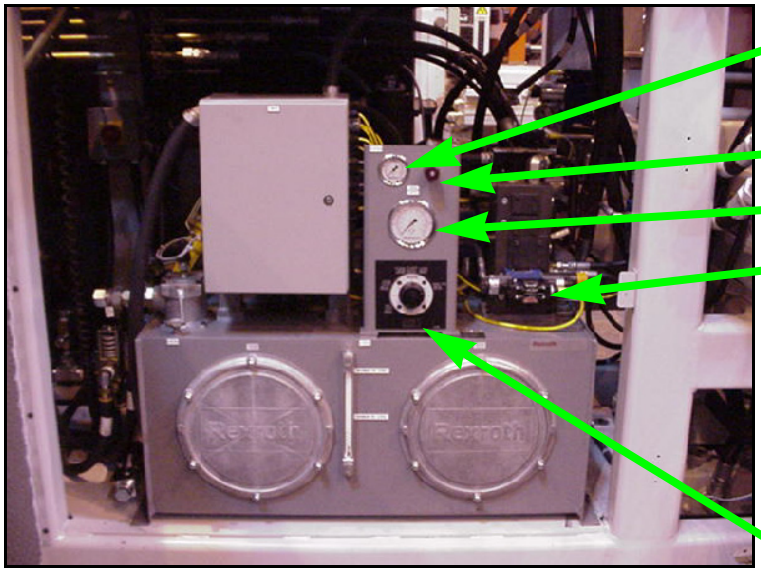
d. Turn  On.

e. On the hydraulic power unit control panel, turn the Clamp/Rotac Selector to CLAMP/ROTAC.

Note: *When set-up is complete, turn the selector switch to Off to prevent unnecessary usage of the hydraulic gauge.*

f. Set the low pressure valve at the rear of the hydraulic unit manifold to 34.5 bar (500 psi).

g. On the MACHINE SETUP - MACHINE ENABLES screen, toggle HYDRAULIC HIGH PRESSURE to DISABLED.



Hydraulic Power Unit (R2000 shown)

Maintenance Checklists

Daily Maintenance Checklist

Put your initials and date of maintenance in the box.

Daily Maintenance Checklist	Week Of: _____/_____/_____						
Procedures	**Sun**	**Mon**	**Tues**	**Wed**	**Thurs**	**Fri**	**Sat**
EXTRUDER							
Safety guarding system: Inspect extruder guarding system.							
Zerk fittings: Lubricate thrust bearing shoes zerk.							
Running temperatures: Adjust.							
Automatic lubrication system (optional); Check low grease level warning light.							
CLAMPING ASSEMBLY							
Safety guarding system: Inspect clamp guarding system.							
Running temperatures: Adjust chilled water flow control valve.							
Automatic lubrication system (optional): Check low grease level warning light.							
Zerk fittings: Lubricate clamp toggle pins, clamp platen shoes, and clamp tie bar bushings.							
TRIMMER							
Zerk fittings: Infeed unit idler gears Outfeed unit idler gears							
Safety guarding system: Inspect safety interlocks							
Machine Safety Circuits							
Test machine safety circuits - hydraulic/ pneumatic and electrical.							

Weekly Maintenance Checklist

Put your initials and date of maintenance in the box.

Weekly Maintenance Checklist	Week Of: _____/_____/_____						
Procedures	**Sun**	**Mon**	**Tues**	**Wed**	**Thurs**	**Fri**	**Sat**
EXTRUDER							
Zerk fittings: Lubricate thrust bearing and extruder drive motor, main hydraulic pump motor, and hydraulic cooling motor.							
Fluid levels: Check reducer gear oil and hydraulic fluid.							
Ball spline: Lubricate.							
Drive belt tension: Check belt tension, adjust if necessary.							
Extruder drawer magnet: Clean magnet.							
CLAMPING ASSEMBLY							
Fluid levels: Check hydraulic fluid.							
Molds: Inspect for missing hardware, grease, dirt, and leaking hoses.							
Zerk fittings: Lubricate main hydraulic pump motor and hydraulic cooling motor.							

Monthly Maintenance Checklist

Put your initials and date of maintenance in the box.

Monthly Maintenance Checklist	Month Of: _____/_____/_____					
Procedures	**Jan**	**Feb**	**Mar**	**April**	**May**	**Jun**
Check all hoses for: Leaks at hose fitting or in hose. Damaged, cut, or abraded cover. Exposed reinforcement. Kinked, crushed, flattened, or twisted hose. Hard, stiff, heat cracked, or charred hose. Blistered, soft, degraded, or loose cover. Cracked, damaged, or badly corroded fittings. Fitting slippage on hose. Other signs of significant deterioration. If a hose is found to be worn or damaged, it must be replaced with a new hose recommended by Uniloy Milacron. Do not use a hose that is not intended for use in the specific application involved. Do not try to repair a hose or a hose fitting.						
EXTRUDER						
Extruder speed reducer (Falk Only): Lubricate input shaft seal and output shaft seal.						
Extruder Heater bands: Inspect extruder and barrel heater.						
Cooling fans: Inspect and retighten as necessary.						
Air filters: Replace filter cartridges.						
CLAMPING ASSEMBLY						
Heater bands: Inspect die head heater bands						
TRIMMER						
Trimmer gear box: Check fluid level. Fill as required.						
Drive belts: Inspect all drive belts. Replace if needed.						

Quarterly Maintenance Checklist

Put your initials and date of maintenance in the box.

Quarterly Maintenance Checklist	Quarterly Of: _____/_____/_____			
Procedures	**1st Qtr.**	**2nd Qtr.**	**3rd Qtr.**	**4th Qtr.**
EXTRUDER				
Repack the thrust bearing: Remove, repack, and reinstall the thrust bearing.				
Accumulator charge pressure: Apply pressure test gauge to nitrogen fill valve.				
Extruder motor: Inspect and maintain using motor manufacturer's manual.				
Extruder speed reducer oil: Flush with 50W oil and replace with an approved oil.				
TRIMMER				
Belt tension: Check tension on all drive belts. Replace if necessary.				
Idler belt pulley: Check idler pulley bearings for wear.				
Tail detabber spring: Check spring wear in tail detabber bars. Replace if necessary.				

Semi-Annual Maintenance Checklist

Put your initials and date of maintenance in the box.

Semi-Annual Maintenance Checklist	Semi-Annual Of: _____/_____/_____					
Procedures	**Date**	**Init.**	**Date**	**Init.**	**Dat**	**Init.**
EXTRUDER						
Extruder drive motor controller: Inspect the drive motor controller.						
Extruder drive motor bearings: Lubricate.						
Hydraulic oil reservoir fluid: Change fluid.						

Semi-Annual Maintenance Checklist	Semi-Annual Of: _____/_____/_____					
Procedures	**Date**	**Init.**	**Date**	**Init.**	**Dat**	**Init.**
Hydraulic power unit: Lubricate per motor manufacturer's instructions, change filters, and clean heat exchanger.						
CLAMPING ASSEMBLY						
Hydraulic oil reservoir fluid: Change fluid.						
Service hydraulic power unit: Lubricate motor per manufacturer's instructions.						

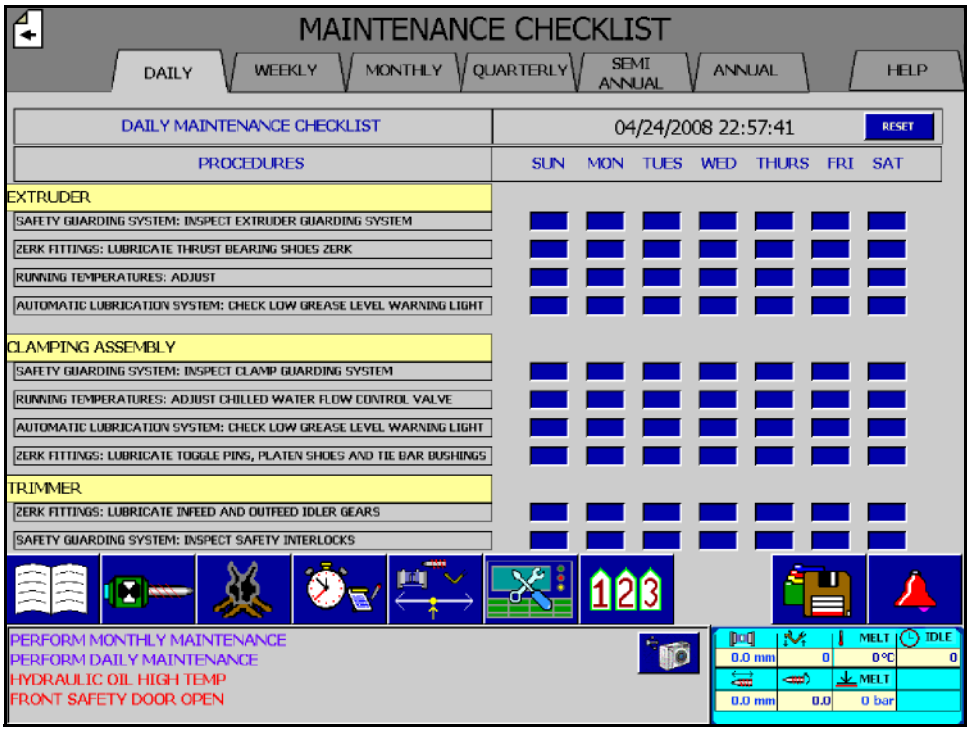
Annual Maintenance Checklist

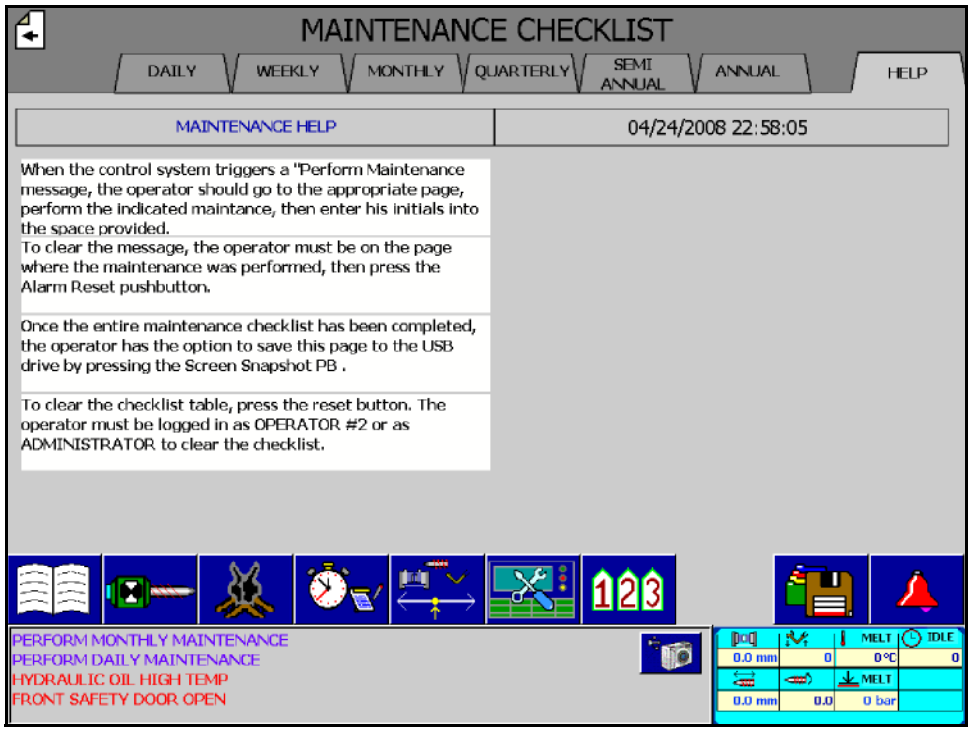
Put your initials and date of maintenance in the box.

Annual Maintenance Checklist	Annual Of: _____/_____/_____					
Procedures	**Date**	**Init.**	**Date**	**Init.**	**Date**	**Init.**
TRIMMER						
Infeed unit: Check wear on infeed unit bearings.						
MACHINE						
Electrical cabinets: Vacuum interior, check wiring connections.						

Maintenance Checklist Screens (Optional, on B&R Control)

The following Maintenance Checklist Screens can be used to check the maintenance tasks that need to be performed on a daily, weekly, monthly, quarterly, semi-annually or annually. The Help pick is very useful when the operator is having a specific problem.





Changing Colors

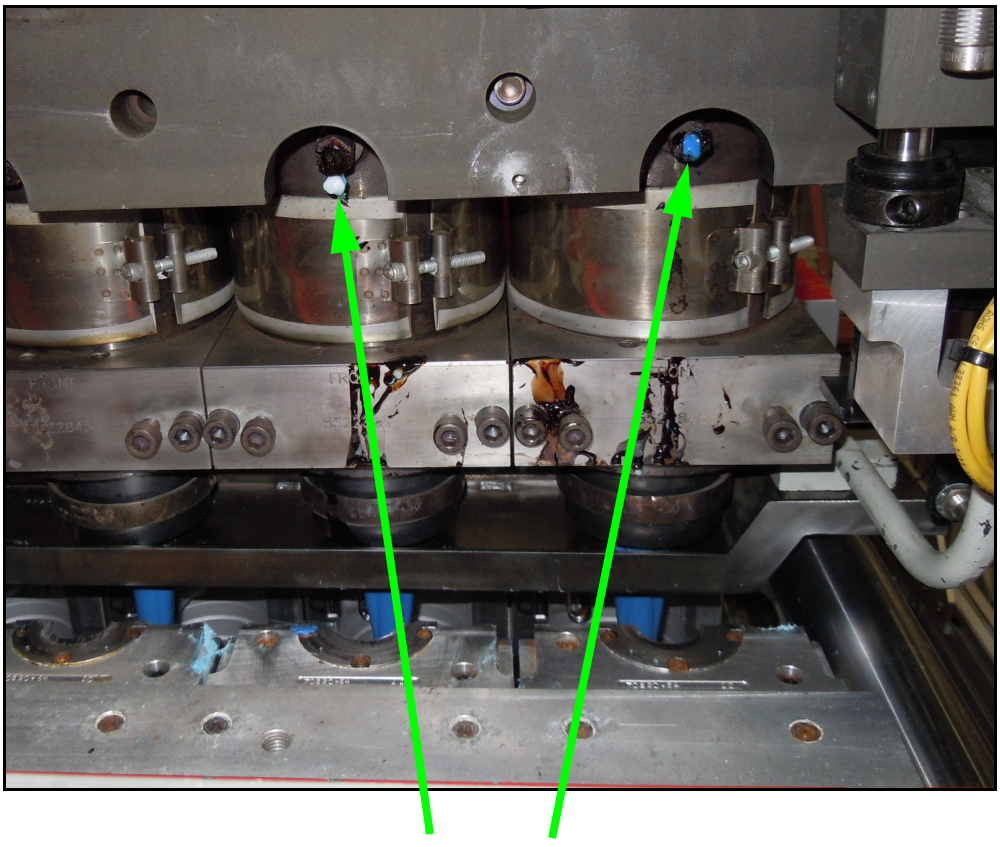
Changing colors on a reciprocating blow molding machine is a straightforward process. By using the procedure below the changeover time can be minimized.

The goal should be to continue to produce good bottles during the color changeover period. This will keep the entire molding process moving efficiently, and will allow the bottles to be reground for use the next time color is changed.

The following procedure should be used when switching resins to a new color.

Color Change Procedure

a. Run the extruder until it is empty of the old color.
b. Raise the head and die heats to 410ºF (210°C.)
c. Remove and clean all old resin from the material handling system.
d. Add the new color to the material handling system.
e. Open the bleed ports on the die head. Turn counter clockwise to open. Start by opening 1/2 turn.



Bleeder Valves

f. Place a catch plate below the bleeder valves to catch the molten resin that exits. The catch plate should be approximately 6” (150 mm) wide by 6’ (1.8 m) long. Aluminum is the best materal for the catch plate.

g. Resume producing bottles with the new resin color. The bottles will be the old color or a mixed color until the old resin is purged from the die head.

h. Adjust each bleed port as needed until approximately 10 mm of material extrudes from each bleed port with each shot. If too much material comes out of the bleed ports, the parisons will not form correctly. Make sure the resin is being caught on the catch plate.

i. Adjust the shot size, if needed, to account for the material coming from the bleeder valves. The goal is to continue to produce good bottles.

j. Continue to cycle the machine until the color of the parisons changes to the new color. The color change will be seen on the center parisons first. When all trace of the previous color is gone from a parison, the bleed port for that parison can be closed. If the previous color returns to the weld line of a parison, the bleed port should be opened again on that parison.

k. As soon as some of the bottles with the new color become acceptable, (normally about 20 minutes of production time), the head and die heats should be returned to normal production temperatures.

l. Continue running the machine and closing the bleeder valves as the parisons take on the new color. The outside parisons will be the last to complete the color change. This process can take up to one hour.